

Break a Gene

Working out why the consequence of a point mutation depends less on the base that changed than on *where in the codon* it sat.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/break-a-gene/

1 Before you open anything

Answer from what you already know. The number does not have to be right — it has to be on paper, so that later you can see exactly which assumption moved.

PREDICTION

One base is changed at random somewhere in a protein-coding sequence. Out of 100 such changes, how many do you expect to leave the amino acid sequence of the protein *exactly* as it was?

Write one or two sentences saying how you arrived at that number.

2 Reading one codon three ways

Open the demonstration. The sequence is the first 30 codons of human *HBB*, the β -globin gene. Each box is one codon, numbered above it: the top row is the coding strand of the DNA, the middle row the mRNA, the bottom row the amino acid. Clicking a DNA base changes it. Clicking the same base again cycles to the next possible base, and once more returns it to wild type.

1. In **Make a mutation**, leave the type set to **Substitution**. Press **Wild type** to clear anything already applied.
2. With no mutation applied, read the **Peptide length** and **Residues altered** readouts. Write them here: _____ aa, _____ altered.
3. Find the box labelled **4**. Its DNA reads C T G. Click the *third* letter, the G. Fill in the first row of the table from the **This mutation** panel.
4. Click that same G twice more, filling in a row after each click.

5. Now click the *second* letter of codon 4, the T. (This replaces the previous mutation — only one is applied at a time.) Click it three times in all, filling in a row after each click.

Base clicked	What changed in the DNA	Codon and amino acid	Consequence	Residues altered
3rd base, click 1				
3rd base, click 2				
3rd base, click 3				
2nd base, click 1				
2nd base, click 2				
2nd base, click 3				

a. Six mutations. In every one of them the DNA changed and the mRNA changed. In how many did the *protein* change?

b. The same codon, the same kind of change, the same number of bases altered. Name the one thing that differed between the top three rows and the bottom three.

3 Is codon 4 special?

One codon is an anecdote. Vary the codon and see whether the pattern holds. The panel titled **Codon *n* — all nine possible substitutions** shows every single-base change to whichever codon you last clicked, sorted into three rows, and each row is labelled with how many of its three options are silent.

1. Click any base inside codon 4. The panel retitles itself **Codon 4**.
2. Below the codon table, the caption begins **Codon 4 of the mRNA is** Take the wild-type codon and amino acid from there, not from the sequence — the sequence is showing your mutation.
3. Copy the three row labels (**1st base**, **2nd base**, **3rd base**) into the table as a count out of 3. A row reading “no silent option” is 0.
4. Repeat for codons 7, 11, 16, 20 and 25.

Codon	mRNA codon	Amino acid	Silent options at 1st base	at 2nd base	at 3rd base
4					
7					
11					
16					
20					
25					

a. One of the three right-hand columns holds the same value in all six rows. Which column, and what value?

b. The third-base column is not constant. Sort your six codons into those with 3 silent third-base options and those with fewer. Then switch to the **Codon table** tab and look at the row labels printed in green. What do the codons in your first group have in common?

Now take it off six codons and onto the whole sequence. Switch the stage to the **Mutation spectrum** tab and press **Run 500 more** twice.

Outcome	Share (your sample)	Exact (all 270 substitutions)
Silent		
Missense		
Nonsense		

c. The caption under the stacked bars gives two numbers directly: the percentage of second-base substitutions that are silent, and the percentage of third-base substitutions that are silent. Write both.

2nd base: _____ % silent 3rd base: _____ % silent

d. State the rule in your own words: what does the position of a substitution within its codon determine?

e. Go back to your prediction in part 1. Compare it with the *Exact* silent figure. If it was off, say specifically what you had assumed that turned out not to hold.

The *Exact* column is not a target typed in by hand. There are 90 bases and three alternatives for each, so 270 substitutions exist, and the page classifies every one of them with the same 64-codon table it uses everywhere else. Your sampled shares have to converge on those values, and they do.

4 Where the shortcut breaks, and how to work backwards

"Third-base changes are silent" is a good generalisation and a bad law. An exam question will hand you the exception, or hand you the consequence and expect you to find the change that caused it.

1. Click any base inside codon 16. Its DNA reads T G G.
2. Read the **3rd base** row of the **Codon 16** panel. Write down all three chips exactly as they appear (codon and amino acid), and the row label.

a. One of those three is red. What consequence class is that, and what happens to translation? Give the length of the peptide it produces.

b. Using the codon table, explain why tryptophan in particular has no silent option at *any* of its three positions, while alanine has three at the third.

Now work backwards from a consequence.

1. Press **Wild type**, then click any base inside codon 7.
2. Switch the stage to the **Codon table** tab. Exactly nine cells are coloured – every codon one base away from codon 7.
3. Exactly one of the nine is red. Click it.

What changed in the DNA	Codon and amino acid	Position hit	Peptide length	Residues altered

c. Only 3.0% of the 270 possible substitutions in this window are nonsense — the rarest of the three outcomes. Explain why it is nevertheless the outcome most likely to destroy the protein, in terms of what the ribosome does when it reaches that codon.

d. Press **The sickle-cell mutation**. Record the base change, the codon change, the amino acid change and the **Residues altered** readout.

This is one missense change out of the 193 possible ones here. Read the **Honest limits** panel, then say in one sentence why the other 192 are not each expected to cause a disease.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it and check yourself — and where you were wrong, write down which step went off.

1. A coding sequence is 90 bases long.

(a) How many different single-base substitutions are possible in it? Show the arithmetic.

(b) Enumerating all of them gives 69 silent changes, of which 66 come from third-base substitutions. Given that a third of the 90 bases sit in each codon position, what percentage of third-base substitutions are silent, and what percentage of *all* substitutions are silent?

2. Not one of the 90 possible second-base substitutions is silent. State what would have to be true of the standard genetic code for a second-base substitution to be silent, and say whether any such case exists. This is an explanation, not a calculation.

3. A single-base substitution changes an mRNA codon from UGG to UGA. This is codon 16 of a 30-codon reading frame.

(a) Which position within the codon was hit, and which base replaced which?

(b) Name the consequence class.

(c) How many amino acids long is the mutant peptide, and how many residues are missing relative to wild type?

4. A classmate argues that deleting three bases from a gene must be about three times as damaging as deleting one. Explain why this is backwards. Your answer should say what a one-base deletion does to the codons downstream of it, what a three-base deletion does to those same codons, and which of the two leaves more of the original protein intact.

6 On your own

Nothing here is graded. The demonstration does considerably more than this worksheet has used.

- Switch the mutation type to **Deletion**, set **How many bases** to 1, and click base 40 (the first base of codon 14). Note **Peptide length** and **Residues altered**. Now set the size to 3 and click the same base. Try sizes 1, 2 and 3 in **Insertion** mode as well. Which number of bases behaves unlike the other two, and why is it the frame rather than the count that matters?
- Press **Clear tally**, then **Run 500 more** one press at a time. How many draws does it take before every value in the *Share* column sits within one percentage point of *Exact*? The triangles under

the bars mark the exact boundaries — watch the bars close on them.

- Mutate a base in codon 1 and read what the page says about the start codon. Then read the **Honest limits** panel to the end and find the one claim it makes about *silent* mutations that this worksheet has quietly ignored throughout.